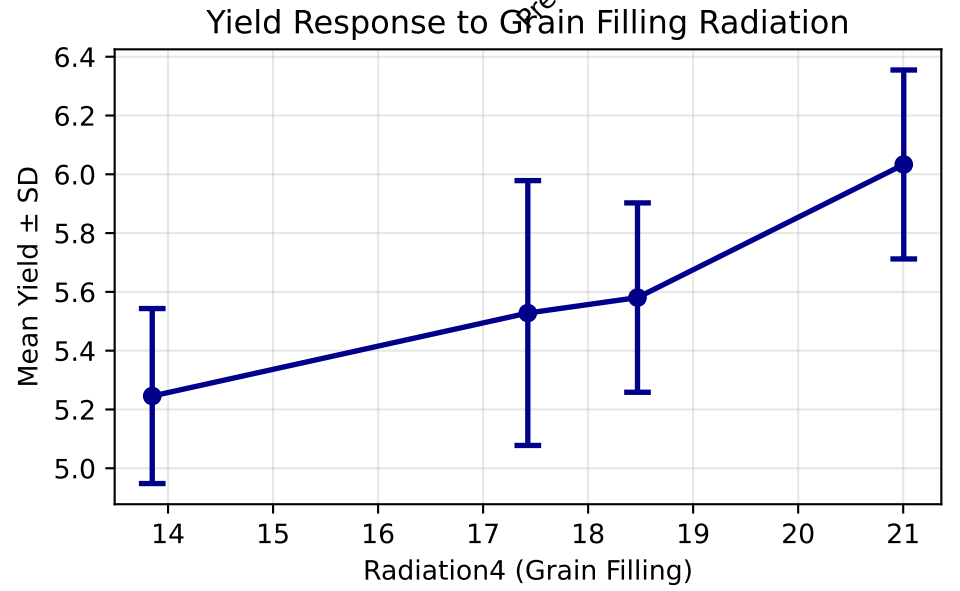
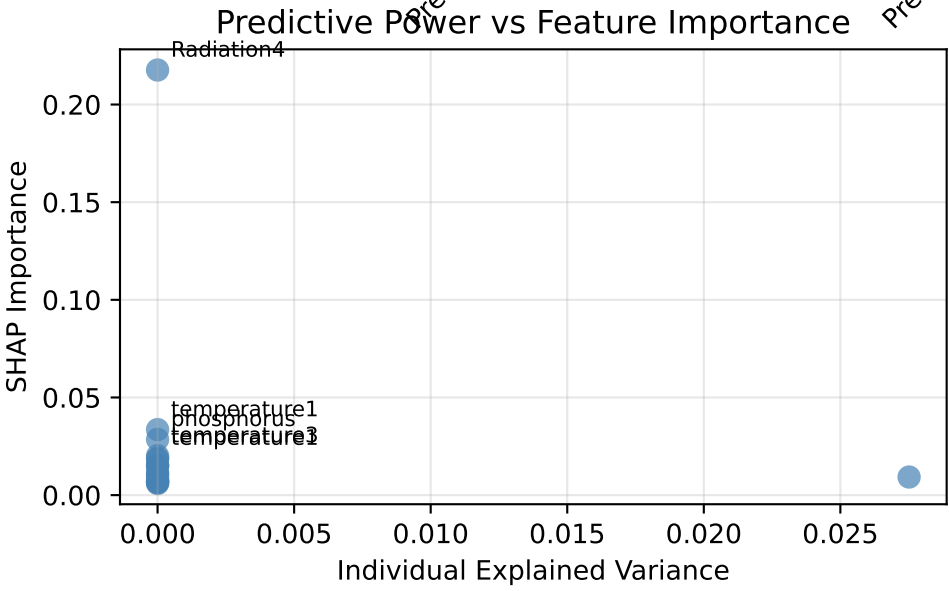
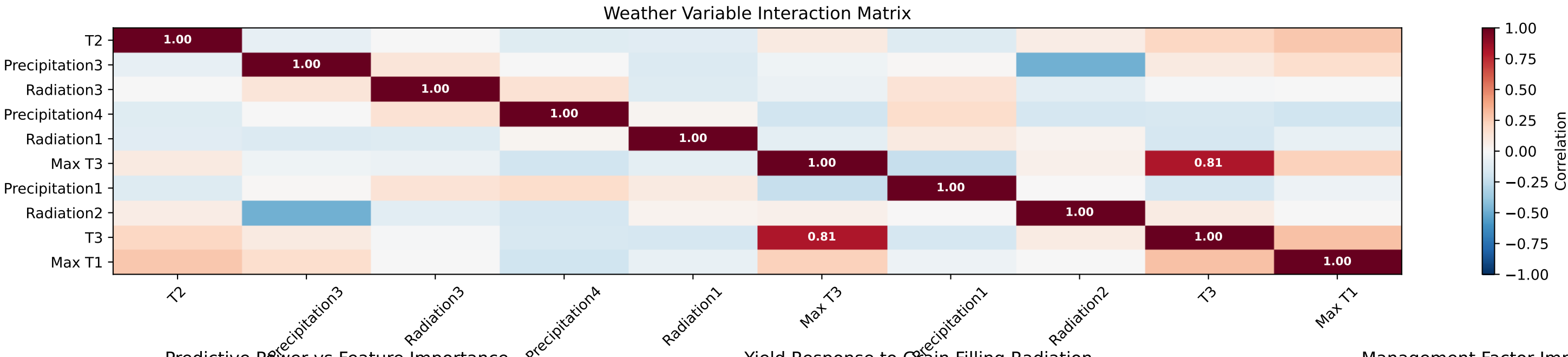
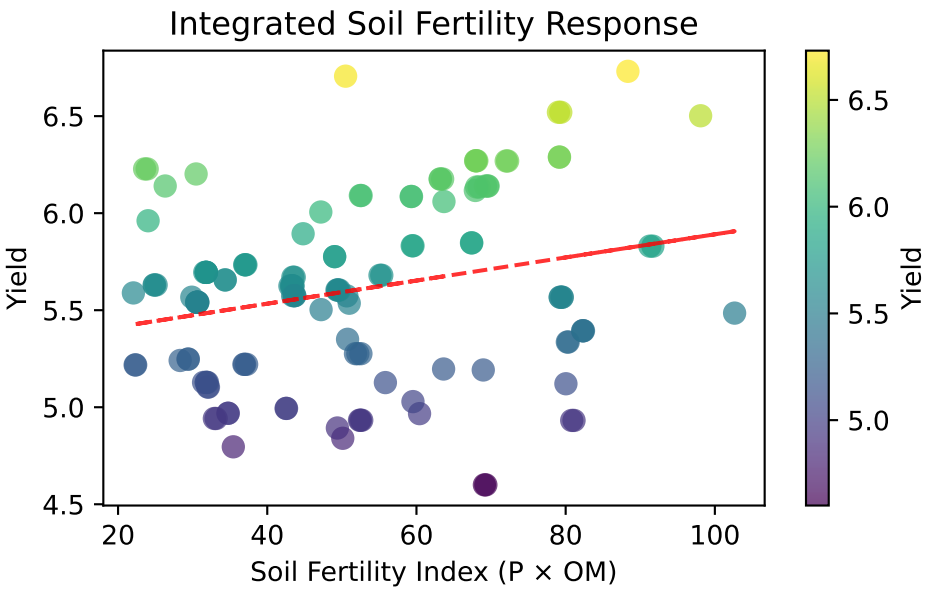
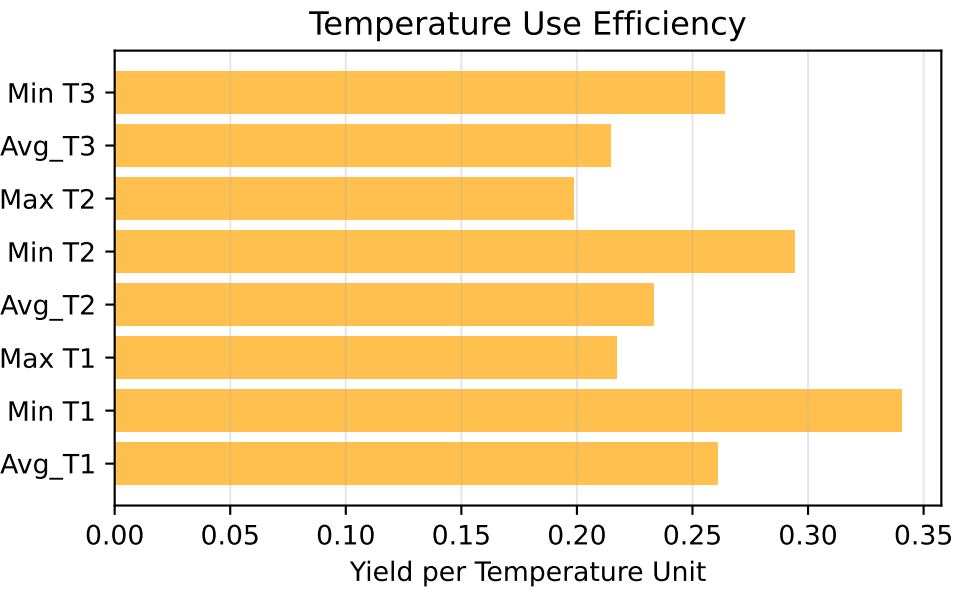
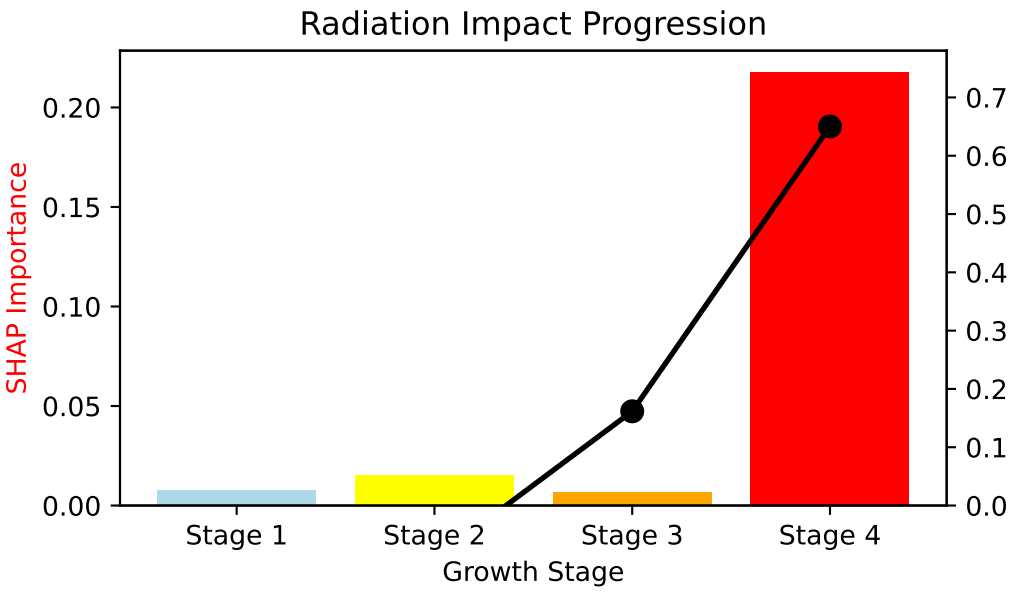
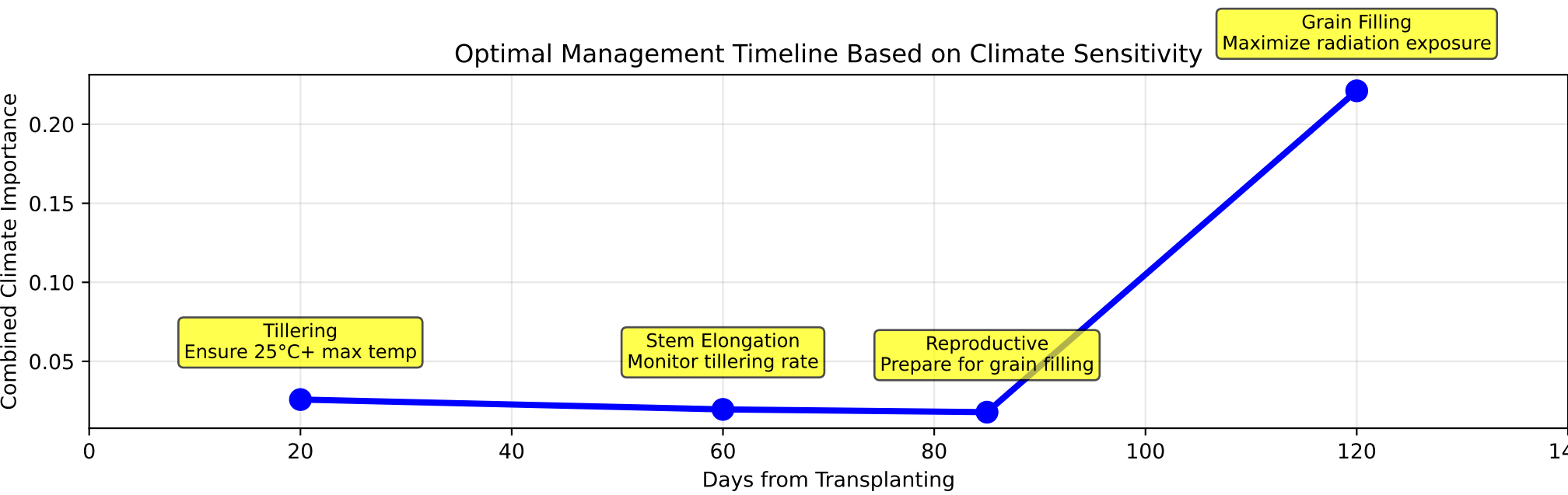
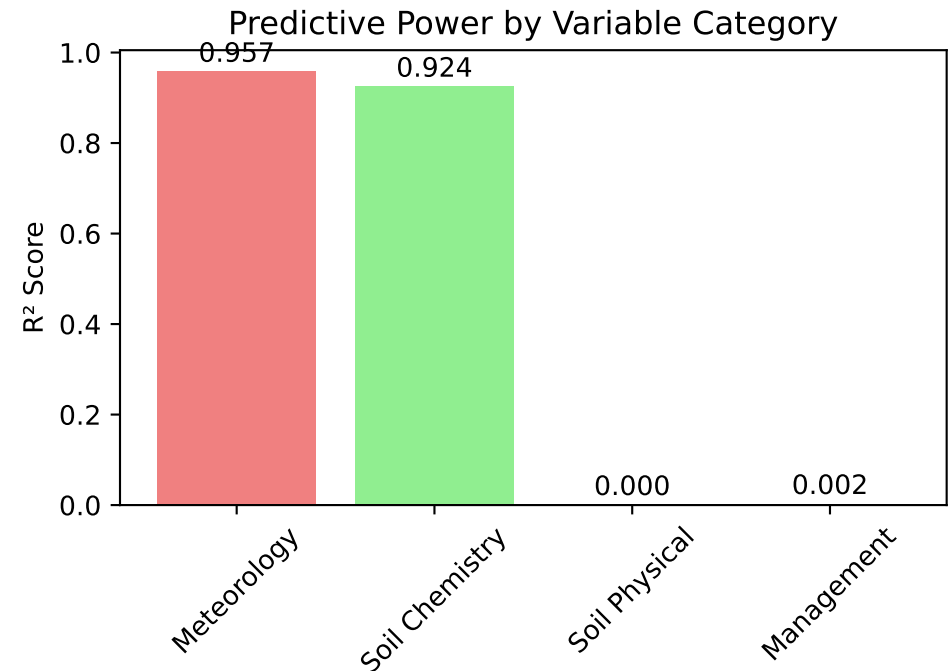
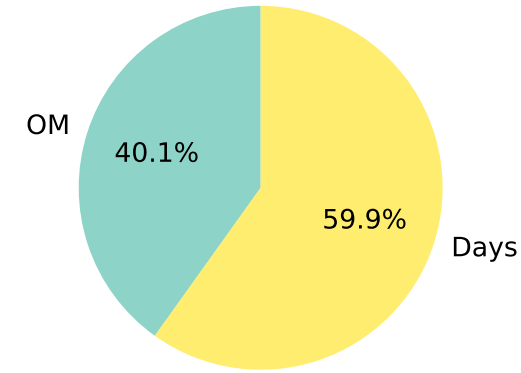


Comprehensive Rice Yield Analysis - Advanced Insights



Management Factor Importance



Rice Yield Optimization - Summary Dashboard

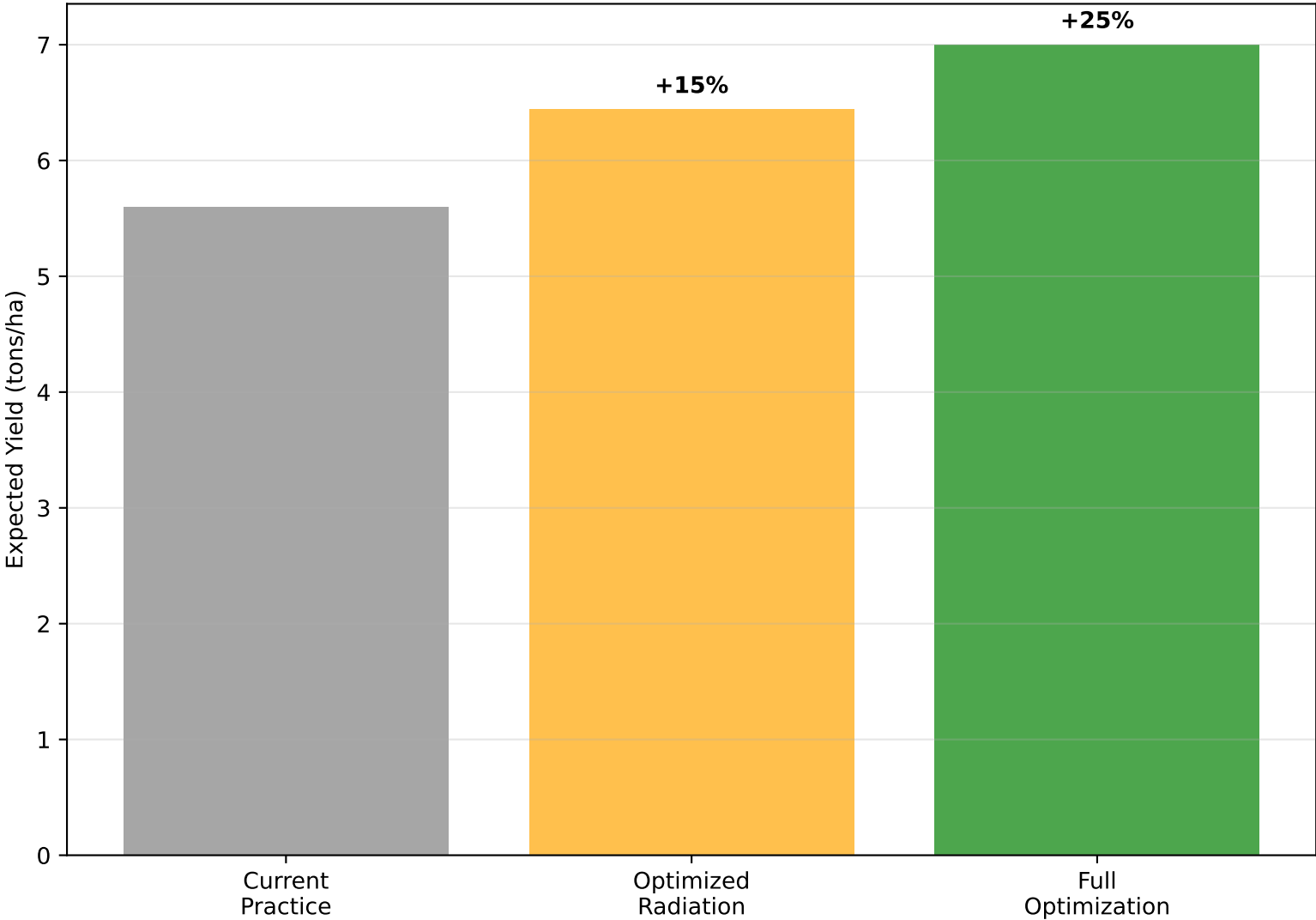
Key Research Findings

Farmer Recommendations

- Radiation4 dominates yield prediction (SHAP: 0.218)
- Early temperature 3× more important than late temperature
- Soil OM optimal range: 2.5-3.5% (correlation effects)
- Weather factors: 54.8% vs Soil: 32.2% importance
- Model explains 66.8% of yield variation
- RMSE: 0.097 tons/ha

1. Time grain filling with peak radiation season
2. Ensure 25°C+ max temps during early growth
3. Maintain soil OM 2.5-3.5% with balanced C:N
4. Apply P early when soil temps rising
5. Use AWD vegetative, flood grain filling
6. Select 110-130 day varieties for timing

Potential Yield Improvements



Risk Assessment for Farmers

- HIGH RISK: Poor radiation during grain filling**
- MEDIUM RISK: Low early-season temperatures**
- MEDIUM RISK: Soil OM outside optimal range**
- LOW RISK: Phosphorus deficiency (manageable)**
- LOW RISK: Precipitation timing (adaptable)**